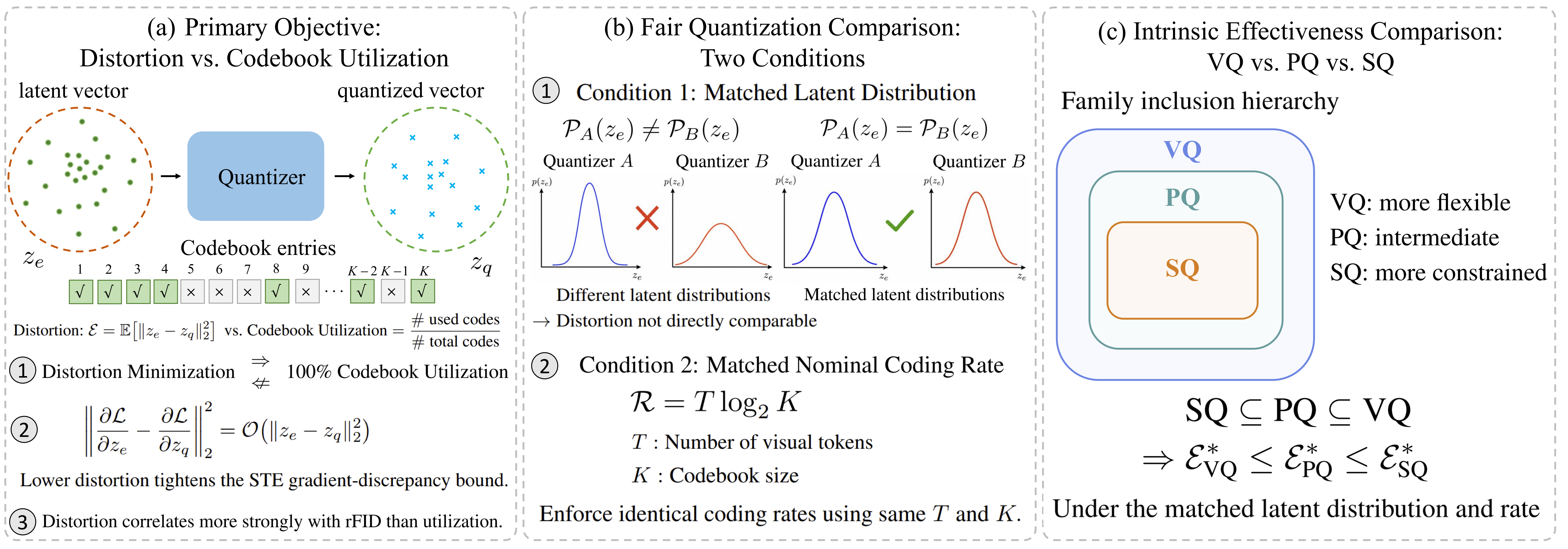

(a) Primary Objective:
Distortion vs. Codebook Utilization
latent vector
quantized vector
Quantizer
z_e
z_q
Codebook entries
1 2 3 4 5 6 7 8 9 $K-2$ $K-1$ K
Distortion: $\mathcal{E} = \mathbb{E}[\|z_e - z_q\|_2^2]$ vs. Codebook Utilization $= \frac{\#\text{ used codes}}{\#\text{ total codes}}$
1 Distortion Minimization $\Rightarrow$ $\not\Leftarrow$ 100% Codebook Utilization
2 $\left\| \frac{\partial \mathcal{L}}{\partial z_e} - \frac{\partial \mathcal{L}}{\partial z_q} \right\|_2^2 = \mathcal{O}(\|z_e - z_q\|_2^2)$
Lower distortion tightens the STE gradient-discrepancy bound.
3 Distortion correlates more strongly with rFID than utilization.
(b) Fair Quantization Comparison:
Two Conditions
1 Condition 1: Matched Latent Distribution
$\mathcal{P}_A(z_e) \neq \mathcal{P}_B(z_e)$
$\mathcal{P}_A(z_e) = \mathcal{P}_B(z_e)$
Quantizer A
Quantizer B
Quantizer A
Quantizer B
Different latent distributions
Matched latent distributions
$\rightarrow$ Distortion not directly comparable
2 Condition 2: Matched Nominal Coding Rate
$\mathcal{R} = T \log_2 K$
T : Number of visual tokens
K : Codebook size
Enforce identical coding rates using same T and K.
(c) Intrinsic Effectiveness Comparison:
VQ vs. PQ vs. SQ
Family inclusion hierarchy
VQ
PQ
SQ
VQ: more flexible
PQ: intermediate
SQ: more constrained
$\text{SQ} \subseteq \text{PQ} \subseteq \text{VQ}$
$\Rightarrow \mathcal{E}^*_{\text{VQ}} \leq \mathcal{E}^*_{\text{PQ}} \leq \mathcal{E}^*_{\text{SQ}}$
Under the matched latent distribution and rate